Module 12: Darwin and Evolution - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Explain natural selection using variation, heritability, and reproductive success.
2. Define fitness in evolutionary terms.
3. Distinguish individual change from population evolution.
4. Use multiple evidence types to support common ancestry.
5. Apply natural selection to a concrete trait example.

Topics

- Historical explanation for adaptation
- Variation and heritability
- Natural selection mechanism
- Evidence for common ancestry
- Fitness as reproductive success

Key Terms to Know

- **Evolution** - Change in heritable traits of populations across generations.
- **Natural selection** - Nonrandom differences in survival or reproduction tied to heritable traits.
- **Adaptation** - Heritable trait shaped by selection in a context.
- **Fitness** - Relative reproductive success.
- **Common ancestry** - Shared origin of lineages through descent.
- **Homology** - Similarity due to shared ancestry.
- **Fossil** - Preserved evidence of past life.
- **Biogeography** - Study of geographic patterns in life.

Core Contents

1. Descent with modification links inherited similarities and differences to shared ancestors. Evolution has no goal or ladder of progress.
2. Natural selection sorts existing heritable variation through unequal reproductive success; need does not create a useful mutation.
3. Fitness is reproductive success relative to others in the same population and environment. Strength or survival alone is insufficient.
4. Fossils, homologous anatomy, biogeography, and DNA support common ancestry. Similar function alone can reflect convergence.
5. Individuals can acclimate; populations evolve across generations. Test selection with evidence of inheritance and reproductive differences.

Study Tips

1. Complete Lab 13 as a prediction, evidence check, and revision exercise. No internet or special equipment is needed.
2. Answer the learning questions as homework practice; show reasoning and units when a question supplies numbers.
3. Use Practice Test 03, then check its self-contained key and explain any correction in your own words.
4. Connect each answer to a mechanism and evidence, rather than a memorized definition alone.

Connected Lab

- `lab-13_darwin-evolution.md`

Generated Visuals

- Module 12: Darwin and Evolution Evidence Map
- Module 12: Darwin and Evolution Reasoning Sequence
- Module 12: Darwin and Evolution Retrieval and Lab Check